

Autodesk RAG App

Husayn El Sharif
Machine Learning Engineer
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LIVE DEMO: <https://huggingface.co/spaces/helsharif/autodesk-rag-assistant>

PORTFOLIO: <https://helsharif.github.io> EMAIL: helsharif@gmail.com

Autodesk Agentic RAG

Hybrid BM25 keyword plus Chroma vector retrieval over an Autodesk corpus, with selectable web-search policy.

☒ Ask ☐ Settings & Eval ☐ Monitoring ☐ About the App

Search mode: Option 2: Local Document Search + Autodesk.com

Local backend: Docling + Chroma + BM25 Hybrid Search

What does AutoCAD LT do?

Press Enter to submit form

Ask

Clear chat



What does AutoCAD LT do?



AutoCAD LT is a precision 2D drafting and documentation software designed for architects, engineers, construction professionals, and designers. It provides best-in-class 2D CAD capabilities but does not include 3D functionalities, which are available in the full AutoCAD version. AutoCAD LT supports some customization through AutoLISP but lacks third-party connections, development tools, and specialized toolsets found in AutoCAD (Local 1, Local 2; Web 2, Web 3).

In summary, AutoCAD LT is a cost-effective solution focused on 2D drafting and documentation tasks, making it suitable for users who do not require 3D modeling or advanced customization features (Web 3).

Answer source: local documents + autodesk.com web search

Search mode: Option 2: Local Document Search + Autodesk.com

Routing note: General product functionality question about AutoCAD LT.

Web search attempted: site:autodesk.com Autodesk What does AutoCAD LT do?

Sources



Outline

01 Problem &
Project Goals

02 End-to-End
System View & Demo

03 RAG Components

04 Key Insights

05 Sample Queries

06 Concluding
Reflections

Problem & Project Goals

- Autodesk customers need accurate, current answers to product questions.
- Build a **Retrieval Augmented Generation (RAG)** Chatbot that can answer typical customer questions from Autodesk **document corpus** & **web search**.

What does **success** look like?

RAGAS METRICS

Faithfulness	>	0.90
Answer Relevance Context Precision Context Recall	>	0.75

BUSINESS KPIs

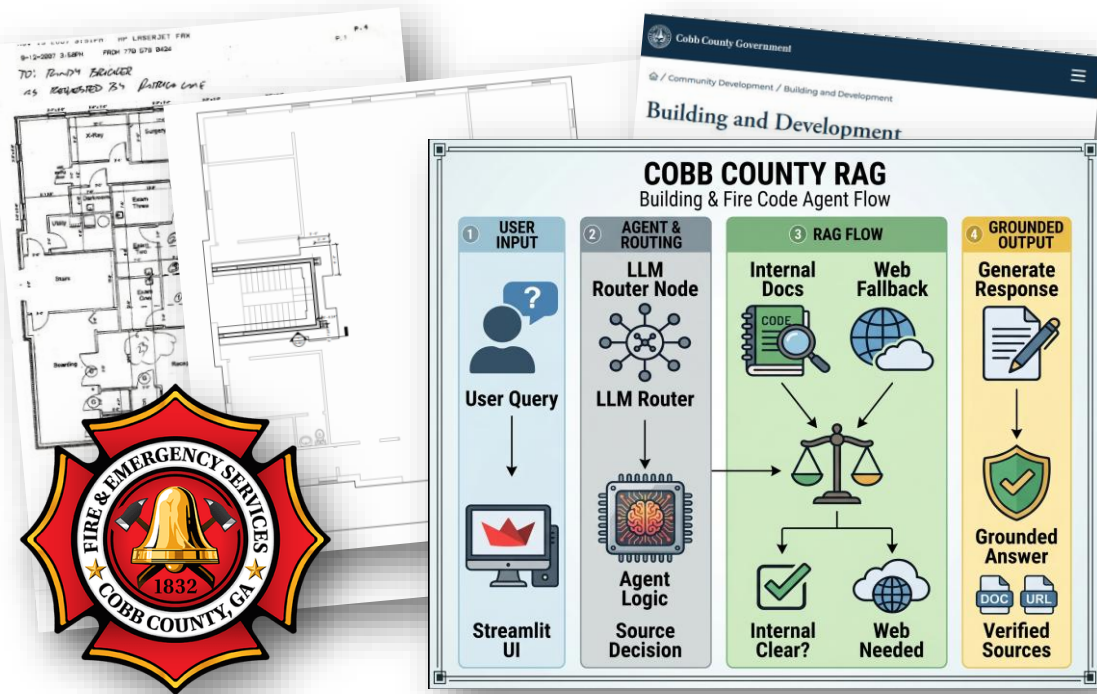
Ticket Deflection/Escalation Rate 

User Satisfaction Score 

Source Click-thru Rate 

Conversion Rate 

Previous Experience



Cobb County Building & Fire Codes RAG

Grounded answers from local PDFs first, with web fallback when retrieval is weak.

☒ Ask ☐ Settings & Eval ☐ About the App

Retrieval backend: Original

Ask about Cobb County building or fire code requirements

Ask

Clear chat

How many parking spots are required for a place of assembly?

For a place of assembly in Cobb County, Georgia, the parking requirement is one space per three seats for every person lawfully permitted within the assembly hall at one time. This applies to assembly halls and similar venues, ensuring adequate parking for attendees during events.

Additionally, for specific types of assembly venues like theaters, the same parking requirement of one space per three seats is also applicable. This standard helps manage the flow of traffic and parking availability during gatherings.

For more detailed information, you can refer to the Cobb County Code of Ordinances, specifically the parking requirements section.

Answer source: local documents

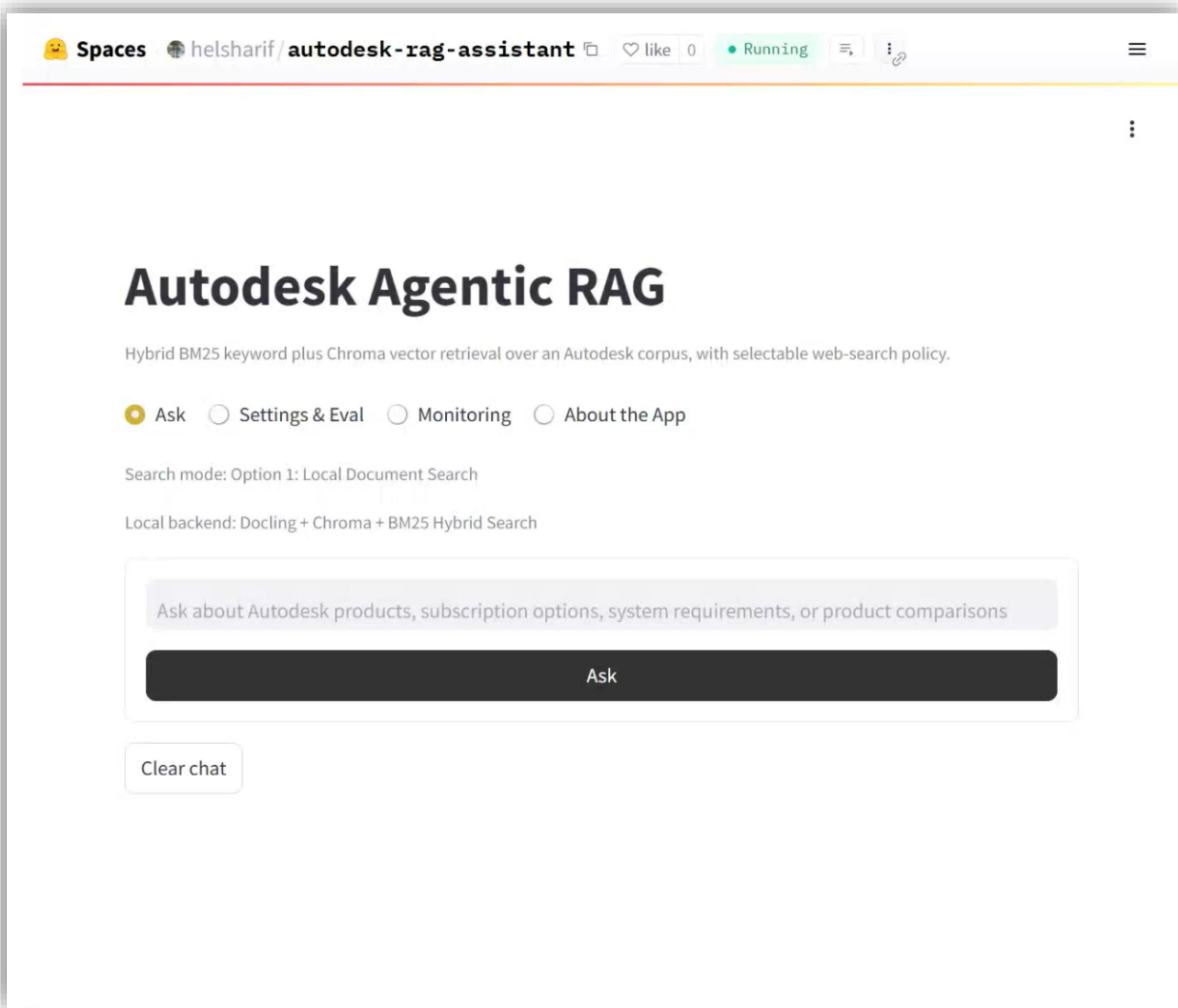
Retrieval backend: Original

Routing note: Local code specifies parking requirements.

Sources

Autodesk RAG App

Live Demo



Autodesk RAG APP

END-TO-END SYSTEM VIEW

Search Modes

	Mode	Web Policy	Best For
1	Local only	No web search	Fast, controlled local answers
2	Local + Autodesk.com	Official Autodesk.com evidence	Current product, plan, version, pricing, support
3	Local + Open Web	Capped open-web evidence	Broader corroboration

Best overall performer

Evaluation Snapshot	Faithfulness	Answer Relevance	Context Precision	Context Recall	Avg Latency
Local Only	0.91	0.66	0.72	0.57	6.3s
Local + Autodesk.com	0.87	0.85	0.81	0.74	12.4s
Local + Open Web	0.84	0.78	0.86	0.62	12.3s

3 Search Mode Selection

Option 1

Local Document Search:
Fastest, local corpus only

Option 2 ★
Local + Autodesk.com:
Preferred default, official current web evidence

Option 3
Local + Open Web:
Broader corroboration, capped for latency/noise

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Local hybrid retrieval backbone

- Chroma semantic vector search
- BM25 keyword search over chunk text + enriched metadata
- Weighted Reciprocal Rank Fusion
- Per-source cap to prevent one document from dominating
- Neighbor chunk expansion: previous + current + next chunk

5

Optional web evidence

Option 2 adds up to 5 official Autodesk.com results with an official source badge.

Option 3 adds up to 3 capped open-web results.

Web snippets become evidence blocks.

6

cross-encoder/ms-marco-MiniLM-L6-v2

Cross-encoder reranker

Local chunks and web snippets are ranked together to keep the strongest evidence blocks.



Evidence sufficient for answering?

Yes

No

Returns a conservative no-answer fallback.

8

Grounded answer output

Concise 2-3 paragraph answer using only supplied evidence with inline source names, local source IDs, or URLs, and no unsupported facts.

User asks Autodesk question:
Natural-language product, pricing, version, workflow, or support question.



Corpus Cleaning & Index Building



Evaluation

50-question golden dataset

LangSmith Experiments

Metrics: Faithfulness, Answer Relevance, Context Precision, Context Recall, Latency

Dashboard tracks status, results, and monitoring logs

EDA & Pre-Processing

The screenshot shows a complex web page with a sidebar on the left containing navigation links and a main content area. The page is cluttered with various elements, including logos (Autodesk, Beautiful Soup, Trafilatura, Lingua), text, and code snippets. The text is a mix of HTML code and rendered content, illustrating the 'Noisy HTML' mentioned in the list.

- Noisy HTML, cleaning & filtering required
- HTML → Markdown (*.md)
- Enrich with metadata, subheadings, TF-IDF keywords

	Raw Corpus	After Cleaning
File Count	1218	974
Corpus Size	303 MB	4.4 MB
Char. count	317 M	3.5 M

Over 98% reduction!

Index Building: Chunking, Tokenization, & Retrieval Artifacts

- Converted 974 cleaned Markdown docs into retrieval ready-chunks
- Doc-aware / heading-aware chunking. 19K+ chunks
- Regex tokenizer for BM25; OpenAI embedding model for Chroma vector index

BM25 Keyword Index

Fast Lexical / Keyword Search

Includes metadata, headings,
TF-IDF keywords

Best for product names, versions, plans,
technical terms

Improves Lexical recall



ChromaDB Vector Index

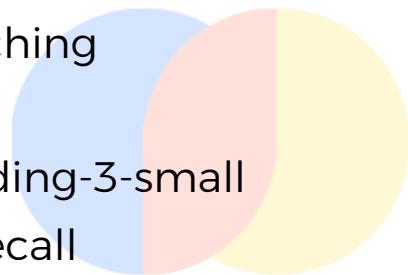
Supports Semantic Vector Search

Meaning-based matching

Dense embeddings

OpenAI: text-embedding-3-small

Improves semantic recall



Search Modes & Hybrid Retrieval

Option	Search Mode	Behavior	Best use
1	Local Document Search 	Uses only local Autodesk corpus chunks from Chroma and BM25.	Fast, controlled local-corpus baseline.
2	Local Document Search + Autodesk.com 	Combines local retrieval with official autodesk.com web evidence.	Preferred mode for current official Autodesk facts.
3	Local Document Search + Open Web 	Combines local retrieval with capped open-web results.	Broader corroboration when official-only search may miss context.

- Hybrid retrieval pairs BM25's fast exact-match search with vector search's deeper semantic understanding.
- Context expansion also adds neighboring chunks from source docs.
- Retrieval results are fused with weighted **Reciprocal Rank Fusion (RRF)**, followed by **cross-encoder reranking** against the user query.

Answer Generation

- The top retrieved evidence is passed through an LLM (GPT-4.1-mini) adequacy gate before generating an answer.
- Only allows answers when the required fact is **explicitly supported by retrieved evidence**.
- Blocks vague or incomplete evidence to prevent unsupported / hallucinated answers.
- Failure returns conservative **no-answer response**.



✓ **PASS**

Generate grounded answer.

⊗ **FAIL**

“I could not find a reliable answer in the available documents or web sources.”

User Interface: Ask

Autodesk Agentic RAG

Hybrid BM25 keyword plus Chroma vector retrieval over an Autodesk corpus, with selectable web-search policy.

☒ Ask ☐ Settings & Eval ☐ Monitoring ☐ About the App

Search mode: Option 1: Local Document Search

Local backend: Docling + Chroma + BM25 Hybrid Search

What's the difference between AutoCAD and Revit? Press Enter to submit form

Ask

Clear chat



What's the difference between AutoCAD and Revit?



The main difference between AutoCAD and Revit lies in their core purposes and functionalities. AutoCAD is a general-purpose CAD software primarily used for creating precise 2D drawings and 3D models. It is a broad drawing tool applicable across various industries for drafting and schematic creation. In contrast, Revit is specialized software for Building Information Modeling (BIM), focusing on creating intelligent 3D models that incorporate detailed building data and support the entire design and construction process (Web 1, Web 2, Web 5).

While AutoCAD is more mature as a drafting package and widely used for detailed 2D drafting, Revit offers a more integrated approach to building design by enabling users to work with a comprehensive 3D model that includes architectural, structural, and MEP components. Both tools can be used together, as they are interoperable, but Revit is generally recommended for those focusing on BIM workflows in architecture and construction (Web 2, Web 3, Web 4).

Answer source: autodesk.com web search

Search mode: Option 2: Local Document Search + Autodesk.com

Routing note: Comparison of Autodesk products based on features.

Web search attempted: site:autodesk.com Autodesk What's the difference between AutoCAD and Revit?

Sources

<https://www.autodesk.com/ca-en/solutions/revit-vs-autocad> (score 10.13)

<https://www.autodesk.com/solutions/revit-vs-autocad> (score 7.99)

<https://www.autodesk.com/support/technical/article/caas/sfdarticles/sfdarticles/Differences-between-AutoCAD-and-Revit-products.html> (score 7.32)

<https://forums.autodesk.com/t5/autocad-architecture-forum/architecture-vs-revit/td-p/5955681> (score 6.50)

<https://forums.autodesk.com/t5/revit-architecture-forum/revit-amp-autocad/td-p/13331217> (score 5.96)

User Interface: Settings & Eval

Autodesk Agentic RAG

Hybrid BM25 keyword plus Chroma vector retrieval over an Autodesk corpus, with selectable web-search policy.

☐ Ask ☒ Settings & Eval ☐ Monitoring ☐ About the App

Settings & Eval

Retrieval configuration

- ☒ Option 1: Local Document Search
- ☐ Option 2: Local Document Search + Autodesk.com
- ☐ Option 3: Local Document Search + Open Web Search

Option 1: Local Document Search uses only indexed local Autodesk corpus documents. The router does not evaluate whether web search is suitable, and web search is disabled.

All three options use the same local Docling + Chroma + BM25 Hybrid Search backend. The radio button controls whether web evidence is disabled, scoped to autodesk.com, or open web.

Evaluation Metrics

Saved metrics load from `eval_results/docling_chroma_bm25_hybrid_autodesk_web_results.json`. Evaluation runs the fixed 50-question `eval_testset/autodesk_testset.csv` dataset in a background process.

Last run: 2026-05-21T18:51:03+00:00 UTC | Questions: 50

LangSmith dataset: autodesk-rag-eval-testset-f4c98b890852

[Open LangSmith experiment](#)

Faithfulness	Answer Relevance	Context Precision	Context Recall	Latency
0.87	0.85	0.81	0.73	13.0 12.4 22.9
✓ Strong	✓ Strong	✓ Strong	• Moderate	• Moderate Avg P50 P99

Evaluation details

Re-run Evaluation

Evaluation details

Rows are sorted to match `eval_testset/autodesk_testset.csv`; the first six rows are the required reviewer questions.

#	Question	App Answer	Reference Answer	Scores
1	What does Fusion 360 do?	Fusion 360 is an integrated platform by Autodesk that combines 3D CAD, CAM, CAE, PCB, data management, and collaboration tools. It enables users to create 3D designs, perform simulations, generate toolpaths for manufacturing, and manage product development workflows all within one software environment. This includes specialized machining strategies such as 4- and 5-axis toolpaths, turning strategies, and stock simulation to verify toolpaths and avoid collisions (Local 1, 2, 3; Web 1, 3). Additionally, Fusion 360 supports simulation capabilities to test designs against various failure criteria and optimize design changes, with cloud-based solving to speed up computations. It also facilitates	According to the document 'Autodesk Fusion Manufacturing Cloud Autodesk Fusion', Autodesk Fusion (formerly known as Autodesk Fusion 360) is design, engineering, electronics, and manufacturing software all-in-one. It connects the entire product development process into one cloud-based platform with integrated 3D CAD, CAM, CAE, and PCB. The document 'Compare Fusion 360 vs Fusion 360 for Personal Use Autodesk' further describes it as	Faithfulness: 1.00 Relevance: 1.00 Precision: 1.00 Recall: 1.00 Latency: 20.1s Evidence: local + web

User Interface: Monitoring

Monitoring

Runtime monitoring summarizes recent RAG interactions logged to Supabase, including retrieval choices, source usage, latency, no-answer outcomes, and errors.

This demo logs submitted questions, generated responses, retrieval metadata, latency, and errors for monitoring and debugging. Do not enter sensitive, private, or confidential information.

Supabase monitoring configured: Yes

Filters

Date range

Backend

Outcome

Used web

No-answer

2026/05/22 – 2026,

All

All

All

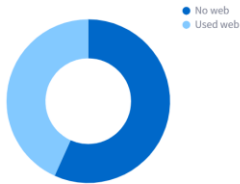
All

Total logged questions	Average latency	P50 latency	P99 latency	No-answer rate
37	12.03s	10.20s	44.98s	21.6%
Web fallback rate	Adequacy pass rate	Error rate	Average source count	Average tokens
43.2%	78.4%	0.0%	3.9	1012
Total estimated cost				
\$0.0170				

Backend usage



Used web vs not used web



Backend	Requests	Percent
Local only	18	48.6%
Local + Autodesk.com	11	29.7%
Local + open web	8	21.6%

Web usage	Requests	Percent
No web	21	56.8%
Used web	16	43.2%

Top Source Documents

Rank	Document Title
1	Autodesk AutoCAD LT 2024 Get Prices & Subscribe To AutoCAD LT 2024
2	Autodesk AutoCAD 2024 Get Prices & Subscribe To AutoCAD 2024
3	CAD Software 2D and 3D Computer-Aided Design Autodesk
4	https://www.autodesk.com/blogs/autocad/autocad-vs-autocad-lt
5	Maya 2024 Features Autodesk

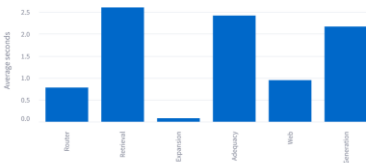
Recent Interactions

created_at	retrieval_backend	router_decision	used_web	adequacy_answerable	no_answer	source_count	latency_t
2026-05-23 20:41:55.7	docling_chroma_b m2s_hybrid_autodesk_web	Comparison of Autodesk products based on features.	True	True	False	5	17.685584
2026-05-23 10:45:19.6	docling_chroma_b m2s_hybrid_local_only	Local-only mode: web search disabled.	False	True	False	5	13.169869

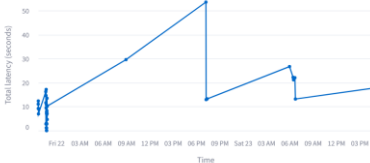
Latency Diagnostics

These charts are more useful than time trends for small datasets because they show where each RAG request spends time.

Average latency by pipeline stage



Total latency over time



Evaluation

- **Golden dataset:** 50 curated Autodesk questions generated from the cleaned corpus using Claude Sonnet 4.6.
- **Question mix:** Simple fact lookups, reasoning questions, and multi-context synthesis questions.
- **Ground truth:** concise, source-grounded answers aligned with the RAG agent's response style.
- **Bias control:** Claude authored the test set; OpenAI generated app answers.
- **Evaluation:** GPT-5.1 judged Faithfulness, Answer Relevance, Context Precision, Context Recall; latency was recorded.

question	ground_truth
What does Fusion 360 do?	According to the document 'Autodesk Fusion Manufacturing Cloud Autodesk Fusion', Autodesk Fusion
What's the difference between AutoCAD and Revit?	Based on the document 'Thomas & Hutton Site Development Drives the Future of Building Design
Does AutoCAD LT do 3D?	No. According to the document 'Autodesk Account Basics Previous Product Versions Available Versions', the
What's the latest release for Maya?	According to the document 'Autodesk Account Basics Previous Product Versions Available Versions', the
Can I use Fusion 360 on a Mac?	Yes. Autodesk's support documentation (autodesk.com) confirms that Fusion 360 (Autodesk Fusion) is
What are the differences between Standard, Premium, and Enterprise?	The document 'Trade in Your Perpetual License Global Promotions Autodesk' states: 'Our Premium plan is
What productivity gain does the AutoCAD Architecture toolset provide?	According to the document 'Benefits of the Architecture Toolset AutoCAD Autodesk', Autodesk
What operating systems does AutoCAD LT support?	According to the document 'Autodesk AutoCAD LT 2024 Get Prices & Subscribe To AutoCAD LT, AutoCAD LT
What was Autodesk's total revenue in Q3 fiscal 2024?	According to the document 'AUTODESK, INC. ANNUAL FISCAL 2024 THIRD QUARTER RESULTS', Autodesk
How many previous versions can an AutoCAD LT subscriber access?	According to the document 'Autodesk AutoCAD LT 2024 Get Prices & Subscribe To AutoCAD LT, an AutoCAD LT
What is Autodesk Flame used for and what platforms does it support?	According to the document 'Autodesk Account Basics Previous Product Versions Available Versions',
When was ArtCAM discontinued by Autodesk?	According to Autodesk's product documentation found in the reviewed corpus, Autodesk ArtCAM was
What operating systems does Autodesk 3ds Max run on?	The document 'Autodesk Account Basics Previous Product Versions Available Versions' confirms that 3DS
What is Autodesk MotionBuilder used for?	The document 'Autodesk Account Basics Previous Product Versions Available Versions' lists
What is Autodesk Inventor used for?	The document 'Autodesk Account Basics Previous Product Versions Available Versions' lists Inventor
What is Autodesk Navisworks used for?	The document 'BIM Coordination & Collaboration Autodesk BIM Collaborate' references Navisworks directly.
What is the Autodesk Education Community?	The document 'Autodesk Account Basics Previous Product Versions Available Versions' references the
What is Autodesk Civil 3D used for?	The document 'Thomas & Hutton Site Development Drives the Future of Building Design Autodesk' provides
What is Revit LT and how does it compare to full Revit?	According to the document 'Autodesk Revit LT Software Get Prices & Buy Official Revit LT 2023', Revit LT is 'a
What does the Autodesk Premium plan offer?	The document 'Trade in Your Perpetual License Global Promotions Autodesk' states: 'Our Premium plan is
What is the Autodesk Vault product used for?	The document 'Autodesk Account Basics Previous Product Versions Available Versions' lists multiple
Why might an architect choose Revit over AutoCAD for a building?	Based on the document 'Thomas & Hutton Site Development Drives the Future of Building Design',
What are the trade-offs between Autodesk Flex tokens and annual?	The document 'Autodesk Account Basics Previous Product Versions Available Versions' confirms that Flex
Why is generative design in Fusion 360 valuable for manufacturers?	The document 'Autodesk Fusion Manufacturing Cloud Autodesk Fusion' describes Fusion's cloud as
How should a small engineering firm decide between individual.	The document 'Trade in Your Perpetual License Global Promotions Autodesk' states that Autodesk
Why is Autodesk Forma most valuable at the earliest stages of a project?	The document 'BIM Coordination & Collaboration Autodesk BIM Collaborate' describes a project lifecycle
What factors should an organization consider when choosing between AutoCAD LT and AutoCAD?	The document 'Trade in Your Perpetual License Global Promotions Autodesk' confirms that the Premium
How does AutoCAD LT differ from full AutoCAD for electrical engineering?	The document 'Maritz Technologies, Inc. AutoCAD Electrical Toolset Autodesk' provides a direct comparison
Why does Autodesk maintain three service tiers (Standard, Premium, Enterprise)?	The document 'Trade in Your Perpetual License' describes Premium as explicitly targeting businesses that

Search Modes

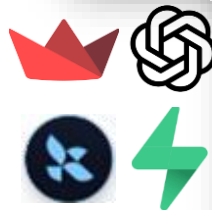
Mode	Web Policy	Best For
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3 Local + Open Web	Capped open-web evidence	Broader corroboration

Best overall performer

Evaluation Snapshot	Faithfulness	Answer Relevance	Context Precision	Context Recall	Avg Latency
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Local + Autodesk.com	0.87	0.85	0.81	0.74	12.4s
Local + Open Web	0.84	0.78	0.86	0.62	12.3s

Deployment & Monitoring

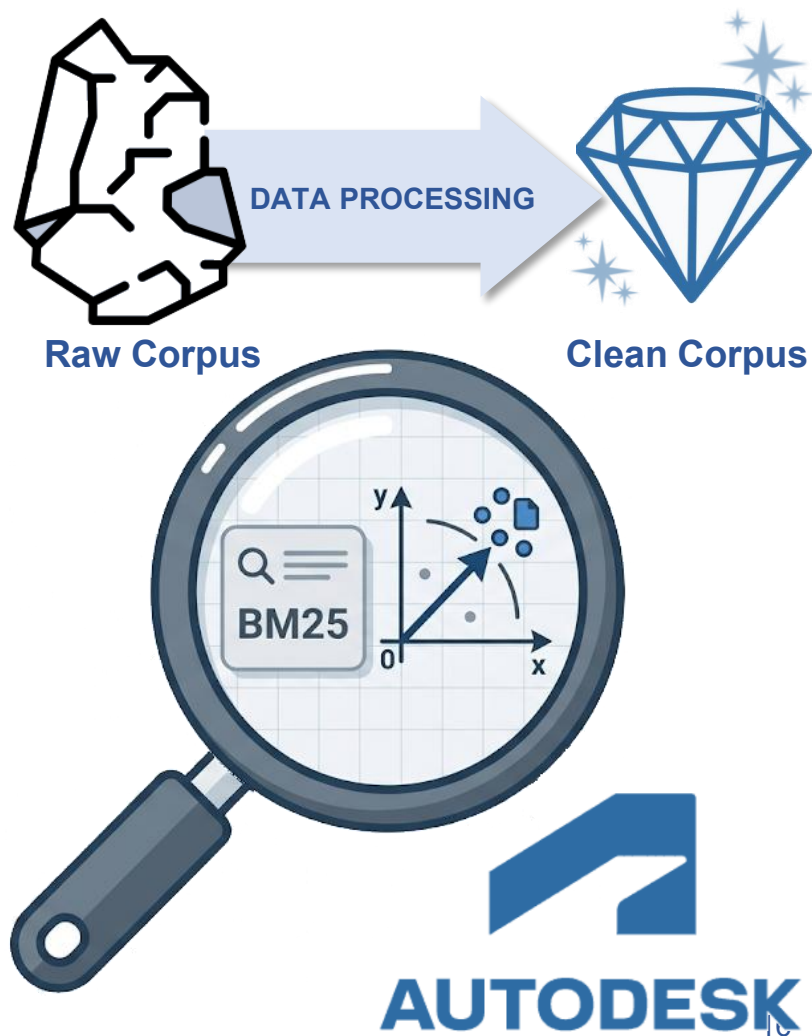
- GIT / GIT Hub (Private Repo)
- Streamlit Application
- Deployed to Hugging Face Spaces (Docker-based)
- Real-time monitoring via Supabase
- API Keys required for LLM, Evals, Open Websearch, LangChain/Langsmith tracing, Supabase stores monitoring logs



A screenshot of the Autodesk Rag Assistant application interface. The top section shows the application title "Autodesk Rag Assistant" and a description: "Hybrid BM25 keyword plus Chroma vector retrieval over an Autodesk corpus, with selectable web-search policy." Below this are tabs for "Ask", "Settings & Eval", "Monitoring", and "About the App". The "Ask" tab is active, showing a search bar and a "Connect" button. Below the search bar is a table with columns "question text", "question_hash text", and "final_response". The table contains several rows of data, including questions like "Which Autodesk product is best for 2D drawing?" and "What is the difference between AutoCAD?". The bottom section of the interface shows a "Project Overview" sidebar with options like "Table Editor", "SQL Editor", "Database", "Authentication", "Storage", "Edge Functions", "Realtime", "Advisors", "Observability", "Logs", "Integrations", and "Project Settings".

Key Insights

- **High quality corpus cleaning** is essential for strong RAG retrieval.
- **Hybrid retrieval** improves coverage across different question types.
- **Official source (Autodesk.com)** web search helps keep answers current.



Key Insights

- **Context expansion + reranking** improve evidence quality before answer generation.
- **Adequacy gating + guardrails** reduce unsupported answers and hallucinations.
- **Evals + Monitoring** make the system inspectable and improvement-ready.
- Proof-of-concept is well positioned for both **technical reliability** and **business value**.



Sample Queries

1. What does Fusion 360 do?

Option 2: Local Document Search + Autodesk.com

Fusion 360 is an integrated platform by Autodesk that combines 3D CAD, CAM, CAE, PCB, data management, and collaboration tools. It enables users to create 3D designs, perform simulations, generate toolpaths for manufacturing, and manage product development workflows all within one software environment. This includes specialized machining strategies such as 4- and 5-axis toolpaths, turning strategies, and stock simulation to verify toolpaths and avoid collisions (Local 1, 2, 3; Web 1, 3).

Additionally, Fusion 360 supports simulation capabilities to test designs against various failure criteria and optimize design changes, with cloud-based solving to speed up computations. It also facilitates manufacturing processes with cutting strategies for different methods like water jet cutting and supports documentation of flat patterns for sheet metal parts (Local 4, 6, 7, 8; Web 4). Overall, Fusion 360 serves as a comprehensive product development platform covering design, engineering, and manufacturing needs.

Sample Queries

2. What's the difference between AutoCAD and Revit?

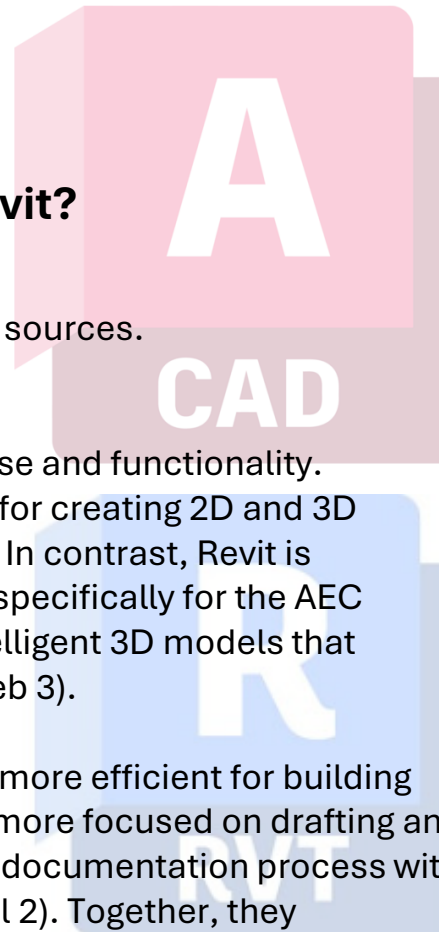
Option 1: Local Document Search

I could not find a reliable answer in the available documents or web sources.

Option 3: Local Document Search + Open Web

The main difference between AutoCAD and Revit is their core purpose and functionality. AutoCAD is a general CAD (Computer-Aided Design) software used for creating 2D and 3D drawings across various industries, serving as a broad drawing tool. In contrast, Revit is specialized BIM (Building Information Modeling) software designed specifically for the AEC (Architecture, Engineering, and Construction) industry to create intelligent 3D models that contain real information about the building components (Web 2, Web 3).

Revit enables faster project creation with integrated data, making it more efficient for building design and documentation workflows. AutoCAD, while versatile, is more focused on drafting and detailing, whereas Revit supports a more comprehensive design-to-documentation process with real-time information embedded in the model (Web 1, Local 1, Local 2). Together, they complement each other by offering effective workflows from design to documentation.



Sample Queries

3. Does AutoCAD LT do 3d?

Option 1: Local Document Search

AutoCAD LT does not support 3D functionalities. It provides precision 2D drafting and documentation capabilities only. In contrast, the full version of AutoCAD includes 3D functionalities such as 3D graphics rotation, advanced geometric views, mesh modeling, surface modeling, and parametric modeling, which are not available in AutoCAD LT [Local 1], [Local 2], [Local 3].

Therefore, if you need to work with 3D models, AutoCAD LT would not be suitable, and you would need the full AutoCAD version.

Sample Queries

4. What's the latest release for Maya?

Option 3: Local Document Search + Open Web

The latest release for Maya is Maya 2026. This version includes new features and updated workflows, and it is accessible through your Autodesk Account (Web 2). Additionally, the 2024 features highlight performance acceleration with interactive workflows and the availability of the Maya USD plug-in as an open-source project (Local 1, Local 2).

Sample Queries

5. Can I use fusion 360 on a Mac?

[Option 2: Local Document Search + Autodesk.com](#)

Yes, you can use Fusion 360 on a Mac. Autodesk provides support for Fusion 360 on Mac systems, and many Mac users have been running Fusion 360 successfully. However, some users report that it can feel somewhat clunky, indicating it may not be as fully optimized as on Windows. Additionally, Autodesk Fusion 360 will require an Apple-supported OS and hardware configuration, especially after updates from March 2025 onward (Web 3, Web 5).

Furthermore, Autodesk also supports running some of its products on Mac through virtualized environments like Parallels Desktop and VMware Fusion, which can be an alternative if you encounter issues with native Mac support (Web 1, Web 4).

Concluding Reflections

What if I had more time?

- Latency
- User-centric features
- Agentic Features
- Business KPIs



Concluding Reflections

Proof-of-Concept to Production

1. Discovery & Product Requirements
2. Scalable Data Ingestion & Processing Pipeline
3. Production RAG Workflow
4. User Interface & Feedback
5. Guardrails & Security
6. Monitoring, Evaluation & CI/CD Pipeline
7. Pilot Deployment
8. Production Launch

Thank you

Husayn El Sharif

Machine Learning Engineer

May 2026

LIVE DEMO: <https://huggingface.co/spaces/helsharif/autodesk-rag-assistant>

PORTFOLIO: <https://helsharif.github.io> EMAIL: helsharif@gmail.com